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import pandas as pd
import matplotlib.pyplot as plt

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans

# Read dataset
data = pd.read_csv("sales_data_sample.csv", encoding="latin1")

# Select numerical columns
X = data[["QUANTITYORDERED", "PRICEEACH", "SALES"]]

# Remove missing values
X = X.dropna()

# Scale the data
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Elbow Method
wcss = []

for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, random_state=42, n_init=10)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

# Plot Elbow graph
plt.plot(range(1, 11), wcss, marker="o")
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.title("Elbow Method")
plt.show()

# Apply K-Means
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
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y_pred = kmeans.fit_predict(X_scaled)


# Display clusters

data = data.loc[X.index].copy()

data["Cluster"] = y_pred


print("Clustered Data:")

print(data[["QUANTITYORDERED", "PRICEEACH", "SALES", "Cluster"]].head(10))


print("\nCluster Counts:")

print(data["Cluster"].value_counts())
```